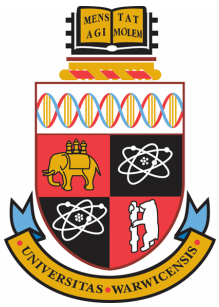
Super-Resolution for Change Detection in Low Resolution Satellite Imagery

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Abstract

Many areas of the world do not have access to accurate maps of buildings. Change detection of buildings is the problem of identifying built or destroyed buildings between two satellite images of the same scene at different times and has uses in urban planning and disaster response. Detecting these precise building changes is difficult with low resolution imagery such as from the satellite constellation Sentinel-2. Sentinel-2 provides access to time series data that has largely been unused for change detection of buildings. Recent deep learning approaches have enabled Sentinel-2 images to be fused in the temporal domain to improve spatial resolution. This project shows that these techniques improve change detection performance for Sentinel-2 imagery, with possible applications for humanitarian mapping.

Keywords: *Deep Learning, Computer Vision, Remote Sensing, Super-Resolution, Change Detection, Satellite Imagery*

1 Introduction

Satellite imagery provides insight into the world at the time it was captured. The field of remote sensing aims to advance techniques used to gain these insights from remotely sensed (satellite) data for use in urban development, poverty mapping, disaster response, and environmental monitoring (1). There is a growing quantity of satellite imagery (2) and remote sensing methods aim to reduce the need for manual interpretation of data, in order to be able to study large areas of the world.

One use for satellite imagery is creating maps remotely. Through the Humanitarian OpenStreetMap Team[1], volunteers can contribute to OpenStreetMap by identifying buildings, roads and other features on high resolution satellite imagery, with a focus on areas of the world where maps are outdated or don't yet exist. The features get added to OpenStreetMap, an extensive vector repository of georeferenced features, that can be accessed by anyone, and used by local governments, communities or humanitarian organisations for making policy decision, urban planning, and delivering aid on the ground.

Buildings and roads are considered two of the most important features to map, and are useful for many humanitarian applications, including aiding development policies, disaster preparedness and disaster response. Building footprints are also used to estimate population size statistics. Accurate statistics, such as population size, are not always available or are outdated for many parts of the world but are needed to inform how humanitarian organisations and governments should act (3).

Building segmentation in remote sensing is the machine learning problem of identifying buildings from satellite imagery, used for example to create maps in areas of the world where they are not readily available (4). A related task is change detection. Change detection in remote sensing is detecting changes between imagery taken at two different points in time (2). At a local scale, when considering urban changes, this gives a map of buildings and roads that have been built or destroyed between the times the two image were taken. Change detection is used for damage assessment after natural disasters (5), tracking deforestation (6), identifying regions of OpenStreetMap that need labelling (7). There are many different sources of remotely sensed data for segmentation and change detection. This project will focus on one satellite imagery source: Sentinel-2, a source that has attracted significant research interest due to its open availability.

1.1. Satellite Imagery

Satellite imagery varies by spatial resolution, revisit frequency, spectral ranges covered (also called spectral resolution), area of coverage, and other factors. A satellite image is simply an array of pixels $I \in \mathbb{R}^{W \times H \times C}$, where W, H, C, are the width, height, and number of channels (also called bands) respectively. Sentinel-2 is a multi-spectral satellite constellation from the Copernicus Programme. The first satellite was launched in 2015, with a second in 2017. Sentinel-2 is unique because of its high spatial resolution compared to other openly available, high revisit frequency satellites (7). It is able to constantly make observations that systematically cover all major land masses, with revisit frequency of 3-5 days for any part of the world. This makes it well suited to collecting time series data. Satellite time series data simply consists of multiple images that represent the same spatial location, defined as $I_{1:N} \in \mathbb{R}^{N \times W \times H \times C}$ where N is the number of time frames in the time series. The imagery produced by Sentinel-2 contains 13 spectral channels: four of which are in the red, green, blue and near-infrared wavelengths, and are captured at a 10 meter resolution, measured in ground sampling distance, or meters per pixel. There are six other bands with a 20 m resolution, and three

[1] https://www.hotosm.org/en/our-approach/

with a 60 m resolution across other wavelengths. Sentinel-2 is typically considered low resolution, compared to high resolution imagery which is in the range of 1 m - 20 cm ground sampling distance.

Sentinel-2 has been well studied in the field of remote sensing, especially for land use mapping and change detection (8). It is well suited for tasks where individual features such as buildings do not need to be identified, and its resolution is sufficient for inferring surface material, reducing processing time costs that come with using high resolution imagery. Mapping features at a fine scale typically requires high resolution imagery (5) (9). High resolution imagery is often distributed from private companies and is costly to obtain (4) (10). There is also a fundamental trade off between spatial and temporal resolution, resulting in high resolution imagery having much longer revisit periods, with potential gaps of 3-6 months between imagery. High frequency observations are useful for conducting studies of change over time (4) and for studying rapidly changing environments. Overall, Sentinel-2 imagery has limited resolving power compared to high resolution imagery - between 10 to 50 times worse, however it has the benefits of being freely available as a high frequency time series. Recent efforts have attempted to use low resolution satellite imagery such as Sentinel-2 for the segmentation and change detection of buildings, either in conjunction with high resolution imagery or with high resolution labels (4) (9) (11).

1.2. Super-Resolution for Change Detection

Whilst the challenge of low resolution imagery initially seems difficult to overcome, super-resolution techniques exist to increase the resolving power of images. For remote sensing, this amounts to increasing the effective ground sampling distance of imagery and many of these techniques have been applied to Sentinel-2 (12). Even more potential for increasing imagery resolution has come from recent works that super-resolve times series imagery together (13). Promising results were achieved by applying these techniques to low resolution imagery for building segmentation (4). However, no known works have extended these results to change detection, which has pointed out as a naturally next step in the same study. Furthermore, few works have investigated change detection on time series imagery, and many change detection models are limited to reasoning over two images rather than time series imagery (8) (11).

This project proposes a naive approach to multi-temporal change detection of buildings: super-resolving time series data from Sentinel-2 into bi-temporal imagery for change detection. Whilst this is effectively just removing the temporal dimension from the problem, this method would make use of existing, but currently not integrated, models in remote sensing, and would show whether useful spatial information is captured by time series data, as suggested by multi-temporal super-resolution methods (13).

2 Background

Research was conducted into related works in super resolution and change detection, particularly into deep learning approaches that deal with time series satellite imagery data, and Sentinel-2 imagery.

2.1. Super-resolution

Super-resolution in computer vision is the task of reconstructing a high resolution image I_{HR} from a low resolution image I_{LR}. Recent techniques use machine learning approaches for this task by learning the inverse of the degradation function f where $f(I_{HR}) = I_{LR}$, to get the reconstruction $f^{-1}(I_{LR})$ (14). The function f can include processes such as downsampling, blurring and adding

noise, which can all be performed with convolution operations. There are several possible degradation function f that can be learnt, making super-resolution an ill-posed problem (14) (13).

Various metrics are used to measure the similarity between the reconstructed image and true high resolution image. Pixel-wise $L1$ or $L2$ loss functions are commonly used. To focus instead on reconstructing higher-level features, perceptual loss is used, where the output of two pre-trained classifiers on the images is compared (14). It is worth noting that resolution increase is not a sufficient metric for comparing super-resolution models, as a simple resolution increase done with bilinear interpolation can create an image of arbitrary resolution.

Results of super-resolution can be improved by utilising time series data (14). The task is the same except now the goal is to reconstruct a high resolution image I_{HR} from time series image data $(I_{LR})_{1:N}$, and this is done by fusing images in the temporal domain (13). There are two intuitive reasons why this would produce higher quality results. Firstly, with a larger sample size it is easier to isolate a signal from noise. Secondly, sub-pixel shifts between images can be exploited for additional detail (13). Sub-pixel shifts occur because of slight variations in where images are taken, so the same 10 m resolution pixel can cover slightly different earth surfaces in each image, assuming images are aligned with pixel-level precision. Both of these are possible because for Sentinel-2 imagery where there is a revisit frequency of 3-5 days, permanent features like buildings change slower than the temporal gap between images, allowing for those features to span most of a time series (4).

Until recently, state of the art time series imagery super-resolution techniques such as DeepSUM and HighRes-net have utilised band-wise super-resolution (14). The DeepSent network improved image quality and spectral consistency for super-resolution of time series Sentinel-2 imagery by fusing band information together before upscaling. A downside of this method is that it concatenates the mean image of the time series with the input during embedding, leading to temporally inconsistent features with any single one of the low resolution images (15). To improve on this, DeepSent was modified to take a leading image from one of the low resolution images, to be the target for super-resolution, instead of using the mean image for concatenation (15). The consistency of the output with the leading image was further improved by re-training the model with a loss function that combines the original L2 loss of DeepSent, with a consistency metric to the leading image.

A different approach upscaled Sentinel-2 imagery as a by-product of learning to segment buildings, by training with a teacher-student model (4). The researchers use HRNet as an encoder, pre-trained on a large RBG image classification dataset, and average the weights across all Sentinel-2 bands, along with extra metadata channels. They modify the encoder blocks to fuse information across times, as opposed to doing 3D convolution, to reduce the number of parameters in the model. Unlike the previously discussed time series super-resolution approaches, they require high resolution imagery for training, rather than using low resolution imagery as the high resolution reference for training and simulating low-resolution imagery from it. They evaluate their model on building segmentation performance and show that the results of super-resolution are better than high resolution imagery downscaled to 4 m ground sampling distance and worse than 3 m ground sampling distance. This provides evidence for DeepSent to super-resolve all Sentinel-2 bands from 10 m up to 3.3 m. In a similar idea to how DeepSent was modified, they experiment with choosing an image to concatenate with features after they are fused. Despite improved performance, they don't find it be be practical as it requires the leading image to be completely cloud free.

2.2. Change Detection

Change detection in remote sensing is the task of segmenting changes between two images of the same scene taken at different times (2). Given two images $I_1, I_2 \in \mathbb{R}^{W \times H \times C}$, the goal is to predict the change map $M \in F^{W \times H}$, where F is a set of classes that describe pixels of interest. In the binary case for building change detection $F = \{0, 1\}$, where 1 indicates a change, such as a building being built or destroyed, and 0 indicates no change between the two images.

Change detection dates back several decades (16). Traditional methods of change detection include pixel-wise difference operations, transformational methods such as principle component analysis, and structural methods such as edge analysis. These simple methods need suitable thresholds to be selected and can struggle with seasonal changes (17). Pixel based approaches also struggle with high resolution imagery (18).

Change detection techniques have followed advancements in the field of computer vision (19) (20), with state of the art methods focusing on supervised deep learning (21). Deep learning methods are useful because they automate feature extraction, generalise well across datasets, and their robustness makes them useful for the varying seasonal conditions seen in satellite data (8). Their generalisability can be seen in the use of transfer learning (1), which helps to alleviate data scarcity, by pre-training on a larger dataset with more available samples. However, these methods can't be used for multi-spectral imagery such as Sentinel-2 (19), since most image datasets contain only three spectral bands. One reason for the improved accuracy of deep learning techniques compared to traditional ones is their ability to learn high level feature representations of real world features (5). Additionally, improvements may be due to the increased availability of computational power and data sources (20) which often limit deep learning approaches (22).

2.2.1. Architectures

Convolutional Neural Networks (CNNs) learn features hierarchically at multiple levels of granularity, allowing for local and global features to be learnt. They improve at making dense predictions compared to super-pixel methods (19), which split images up into groups of pixels to analyse. U-Net is a popular CNN that builds on previous architectures by using a fully convolutional approach (23). In U-Net, images are downsampled with max pool convolutions as they pass through the network, and upsampled in the back end of the architecture with up convolutions, in an encoder-decoder fashion. Skip connections are passed between encoder-decoder layers of the same resolution. They combine detailed information present early in the network with the abstract features present later in the network to preserve local spatial details(19). U-Net was designed for segmentation, a dense prediction task where each pixel in an image is assigned a label (20). The task is highly related to change detection (8), and segmentation methods naturally extend to change detection. Concatenating two images before feeding into a segmentation network can be used to output a change map, and this technique is called early fusion (21) (24). Some change detection methods also utilise multi-task decoders to learn both image segmentation and change detection (11). Alternatively, segmenting images individually before comparing the output, know as post-segmentation, can be used for change detection (5). It should be noted that well performing segmentations methods do not always transfer well to the satellite imagery domain, as changes satellite can be limited to a very small spatial area (i.e. just a few pixels) (25) compared to objects taking up most on of an image in other domains.

FC-EF is an early fusion change detection architecture based on U-Net (19). The authors removed a layer from U-Net, making it more lightweight and combatting over-fitting caused by data scarcity.

As the first fully convolutional architecture for change detection, it outperforms previous models in both change detection performance, and inference speed. Along side it, the authors present a Siamese model that take a middle fusion approach (2). The model, named FC-Siam-diff, uses two separate U-Net encoders with shared weights. Their skip-connections are combined with the difference operation before going to a shared decoder. This difference operation acts as a heuristic to guide the model to detect changes. The FC-EF-Res model builds on FC-EF, by introducing residual blocks, and outperforms FC-EF (20). Residual blocks were introduced to address the degradation problem for image classification, where deeper networks perform worse than shallower ones, but have been shown to improve network performance in both cases (26). Many other architectures have been proposed that are outside the scope of this project that utilise deep learning techniques such as attention mechanisms and transformers (21).

2.2.2. Multi-Temporal Change Detection

Multi-temporal change detection extends the ideas of change detection from two images (called bi-temporal change detection) to time series satellite imagery. The goal of multi-temporal change detection can either be used to locate the point in time when change occurs, or to do change detection between several pairs of images, or simply as an extra modality for bi-temporal change detection.

Continuous Change Detection and Classification (CCDC) is a traditional multi-temporal change detection method. CCDC uses harmonic regression on each pixel to capture seasonal trends, and categorises change as a difference between predicted and actual values that exceeds a threshold. CCDC is implemented in ArcGIS, a popular commercial software that performs satellite imagery change detection and can take Sentinel-2 imagery as an input (27).

Because building change detection is often done with high resolution imagery and time series data is rarely available for high resolution imagery due to long revisit periods, image based multi-temporal change detection for high fidelity features has not been seriously studied until recently (28), and is still a challenging area of research in change detection (21) (8). Whilst post-segmentation can be extended to multi-temporal change detection, these model can't learn temporal relationships, and are outperformed by more recent models, such as MultiUTAE (28). MultiUTAE is based on a U-Net architecture with a temporal attention mechanism. Since the method was proposed in 2024, new architectures have improved on MultiUTAE that are outside the scope of this project (11).

2.2.3. Datasets

Many open datasets exists for change detections and related problems, made to combat the issue of data scarcity for urban satellite image analysis (8) and because large amounts of data is required to train deep learning models (16). For the purposes of this project, a dataset with the following would be required: time series Sentinel-2 imagery, bi-temporal building labels or a building change map. No known dataset exactly contains these two features. There are many high-resolution bi-temporal change detection, but all of them contain imagery taken from before 2015 and so can't be compared to Sentinel-2 imagery. The next best suitable datasets will be discussed here.

The Onera Satellite Change Detection (OSCD) dataset (16) contains pairs of Sentinel-2 images and manually annotated change maps documenting urban growth . It contains 24 urban locations around the world of usually densely populated areas. Over-fitting is an issue with OSCD due to its small size (20). Additionally, it has a class imbalance with many more change pixels compared to no change pixels. Class imbalance is common in most datasets due to the small spatial scale of most

urban changes, and dealing with it effectively is an ongoing challenge for change detection methods (22).

The Fine Urban Change Segmentation (FUSU) dataset (29) is a multi-temporal, multi-modal, fine-grained change detection dataset. Multi-modal change detection utilises imagery from varying sources. Fine-grained change detection classifies pixels into several classes, instead of just one. It contains monthly Sentinel-2 imagery, as well as bi-temporal 30 cm resolution Google Earth imagery of the same location. The high resolution image pairs have pixel annotations with 17 land use classes ranging from traffic land to grass. Although it is localized to urban areas in China, it has pronounced domain shifts due to geographical variation across the country, which is desirably for model generalisability.

The SpaceNet 7 dataset (9) was made for the SpaceNet 7 building localisation and tracking challenge. It contains 60 publicly available labelled areas around the world. Each contains monthly 4 m resolution, 3-channel imagery and polygon building labels, acquired by referencing higher resolution imagery after manual building identification. Like the other datasets, it was specifically made for areas undergoing urbanisation. The dataset has been used for change detection purposes (28).

2.2.4. Evaluation Metrics

Binary change detection inherits many of its metrics from the segmentation problem. Pixels can be classified as either changed or not changed, corresponding to positive and negative labels respectively. If change is correctly identified, that is a true positive (TP), and when there is no change but a pixel was classified as changed, that is a false positive (FP). Similarly there can be true negatives (TN), and false negatives (FN). The most commonly used evaluation metrics are Overall Accuracy (OA), Precision ($Prec$), Recall (Rec), F-Score (F_1) and Intersection over Union (IoU). Because classes are imbalanced heavily towards change, they are generally reported for the change (positive) class only (24), which are the definitions given bellow.

$$OA = \frac{TP + TN}{TP + FP + FN + TN}$$

$$Prec = \frac{TP}{TP + FP}$$

$$Rec = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 \times Prec \times Rec}{Prec + Rec}$$

$$IoU = \frac{TP}{TP + FP + FN}$$

Overall accuracy generally evaluates model performance. A better (higher) precision indicates a model made fewer incorrect change classifications. Improved recall score means fewer change areas were missed by the model (24).

3 Methodology

Multi-image super-resolution for change detection was investigated by comparing the performance of change detection models on super-resolved Sentinel-2 imagery against unmodified Sentinel-2 imagery.

To do this, time series imagery was acquired to create a modified version of the OSCD dataset. OSCD was chosen to ensure label alignment with new imagery and to guarantee partial Sentinel-2 data availability, as it uses Sentinel-2 imagery. DeepSent was chosen as the state of the art super-resolution method. FresUnet was chosen as simple baseline, because the code was openly accessible, and because it performs well on both low resolution and high resolution datasets (20).

3.1. Downloading Imagery

OSCD on its own is a bi-temporal dataset, with two images taken roughly 3 years apart, and a binary change map for urban features. The idea of this experiment is to use available satellite data that is temporally close to both images to form a multi-image stack of nine image frames, as recommended for DeepSent (13).

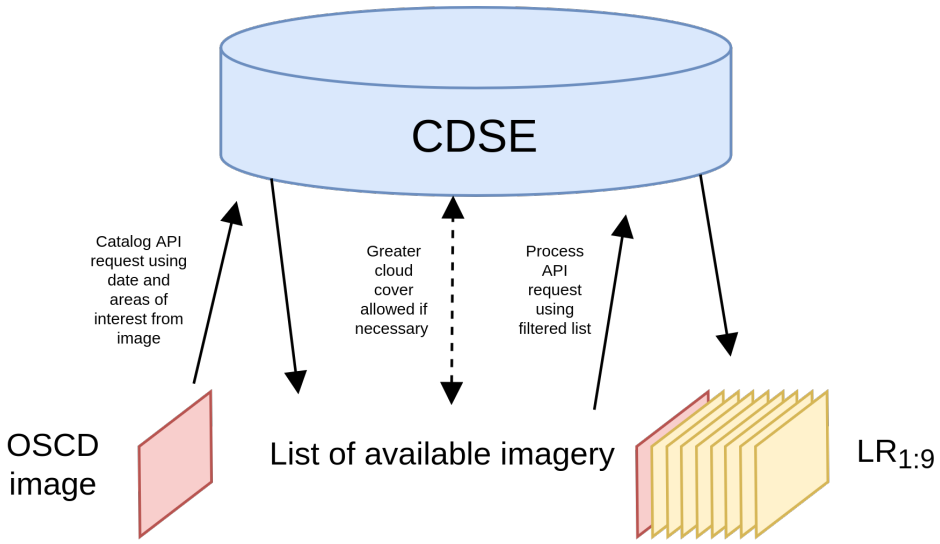


Figure I: The Sentinel-2 imagery download process. A list of available imagery is first requested based on the original OSCD image. If necessary, the list is requested again with a greater maximum cloud coverage allowance. Then this list is filtered down to 9 images to download.

Figure I summarises the data download process. The area of interest and dates provided for each image in OSCD were used to query the Copernicus Data Space Ecosystem (CDSE) Catalog API[2] for available imagery over a time period from the original date up to 600 days later. An issue with using OSCD for this experiment is that the earlier of the two images is captured near the launch of Sentinel-2. This meant imagery exclusively later than the earlier of the two images had to be used to form its multi-image stack. For consistency, both the first and second image stacks are formed in the same way. The Catalog API provides a cloud coverage percentage for returned frames, as well as list of availalbe imagery dates. Only ones with less than 10% cloud coverage were considered, or if less than 9 frames exist, this was extended to 20%, although only one image in the dataset required this. Using this method, stacks span about a year, and in a few cases less than 3 months. It is worth noting that the revisit time of 3-5 days was only achieved two years after the launch of Sentinel-2, meaning for the first image in every pair of the OSCD dataset fewer dates are available than would be expected.

[2]https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Catalog.html

Each frame consists of the Sentinel-2 bands and a mask. Masks were created using scene classification labels computed by CDSE for each pixel, and were set to exclude pixels that were dark, or had no data, and pixels likely to contain clouds, cloud shadow or snow. Often no data pixels are patches of unavailable imagery or bad data from sun flairs reflecting off roofs. The downloaded data was checked for mask coverage, and images with too many excluded pixels were removed from the dataset. Possibly due to changes in CDSE data processing algorithms between the creation of the OSCD dataset and this project, some of the first images in the stack contained a large portion of invalid pixels, even though they would be expected to match the original OSCD dataset images. These and other stacks with a low average number of valid pixels were removed to ensure a full image could be re-constructed using DeepSent. The effects of this on the dataset are shown in Table I.

Stage of OSCD modification		Valid Pixel %	Num samples
	Original OSCD	100.0	24
(1)	Download all images	88.47	24
(2)	Remove low valid pixel samples	95.17	19
(3)	Remove samples unable to be coregistered	94.95	18
(4)	Remove large samples	94.69	17
(5)	Super-resolution	100.0	17

Table I: Different stages of modifying the OSCD dataset. At each stage the percentage of valid pixels is the percentage of pixels not covered by a mask, and the number of samples is the combined test and train count.

3.2. Pre-processing

The most common pre-processing done for change detection is co-registration. Co-registration is the process of aligning multi-temporal images to pixel level precision. Whilst DeepSent co-registers bands together explicitly and images together implicitly (13), image co-registration is still necessary across the two times for change detection (30). This is because images will be overlaid and so must line up pixel to pixel to be meaningful (5). Co-registration was done using the EO-Learn library[3], which uses the ECC algorithm. The two stacks corresponding to a pair of images were all co-registered together. The near-infrared band, band eight, was used as it a high contrast band, and the first image in the first stack was used as the target image for co-registering all 18 images together. The co-registration algorithm of one stack did not converge for several of the images, and this image was removed. Using the same algorithm, better results have been achieved with other parameters (31), however this was not done for this experiment.

The last step of pre-processing was preparing the images for input into DeepSent. Most importantly this involved reverting bands to their original scale. CDSE returns bands crops and upscaled by nearest neighbour 6x, 2x, or 1x to match the highest band resolution. To undo this, images were cropped further to be divisible by 6 and downscaled using nearest neighbour. The dataset made up of the first image in each of the stacks will be hereafter called "OSCD-LR". It is exactly like the original OSCD dataset, but with only 18 samples, slight cropping, and variations in pixel intensities from CDSE processing algorithm differences.

3.3. Super-resolution

[3]https://eo-learn.readthedocs.io/en/latest/



Figure II: Comparison of low-resolution (OSCD-LR) and super-resolved (OSCD-SR) images with the ground truth change map. Taken from a crop of the Saclay West sample.

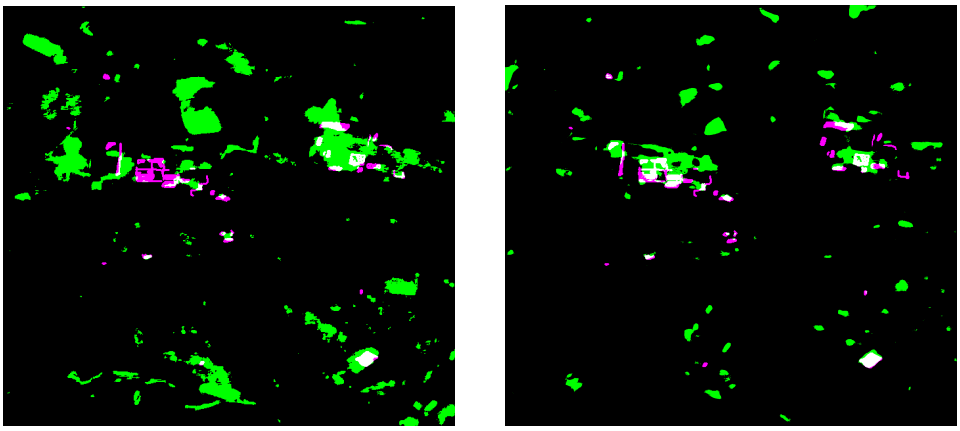
4 Results

Evaluating the super-resolved OSCD dataset (OSCD-SR) was done by training and testing on the FresUnet model (20). FresUnet was chosen for this experiment as the source code and default parameters for training the model are openly available[4], as well as the model showing good performance on OSCD compared to models discussed in Section 2. The shallower version of the network, used by the authors on the OSCD dataset to avoid over-fitting, was employed. As identified by others, OSCD is a small dataset and so models tend to over-fit to it (20), and this was observed in several runs where the best epoch occurred early on. Additionally, maximum accuracy varied between runs. Therefore, metrics from the best epoch across several runs are reported in Table II, determined by the best F-score. As a baseline, the FresUnet model was also run on the original OSCD dataset (OSCD-LR), and on a bi-cubically upscaled version of the OSCD dataset (OSCD-BC), to the same resolution as OSCD-SR. Bi-cubic upscaling is commonly used as a baseline in super-resolution (13). The same hyper-parameters were used, other than a smaller patch size for testing and training OSCD-LR.

The super-resolved dataset improved the change detection performance of FresUnet across all metrics. Accuracy increased the most in the change class, resulting in fewer missed changes and fewer false predictions, reflected in the precision, recall and F1 metrics. These improvements can be seen visually in Figure III. These findings support the claims that DeepSent improves perceptual image quality over unmodified Sentinel-2 imagery (13).

Dataset	Accuracy			Precision	Recall	F-Score
	Global	Change	No Change	Change	Change	Change
OSCD-LR	95.18	41.30	97.85	48.60	41.30	44.65
OSCD-BC	92.24	51.25	94.26	30.57	51.24	38.30
OSCD-SR	95.68	51.15	97.88	54.36	51.15	52.71

Table II: Results of training and testing FresUNet on each of the datasets. All Metrics are in percentages.



(a) OSCD-LR (b) OSCD-SR

Figure III: Results of FresUnet on the two datasets for the Saclay West sample. White pixels are true positives, black pixels are true negatives, green pixels are false positives, pink pixels are false negatives.

[4]https://github.com/rcdaudt/fully_convolutional_change_detection

One factor that limits model performance in this case is the noise introduced from upscaling low resolution labels. An issue with using the OSCD dataset is that upscaling labels introduces noise into the dataset, as they were not made to show changes at a high resolution, and so change pixels may no longer be accurate. Noisy labels have been shown to reduce effective batch size, and to take a larger training set to reach the same prediction accuracy as clean labels (32). OSCD is already a small dataset, and especially after the reductions in size explained in Section 3, these experiments likely can't show if super-resolution can reach high prediction accuracy. The SpaceNet 7 dataset, in contrast, uses georeferenced polygon labels. Whilst these were made mostly using 4 m resolution imagery, they much more precisely outline buildings, and will retain their resolution better when upscaled. Therefore experimenting on this dataset would be interesting for further evaluation.

5 Conclusion

This project investigates using multi-image super-resolution to preprocess Sentinel-2 time-series imagery for change detection of buildings and shows promising results through experiments on the OSCD dataset. This problem is an important area of study in remote sensing because of the open availability of low resolution Sentinel-2 imagery, the need for accurate change detection methods, and to motivate the study of time-series change detection.

Additional challenges of dealing with time series imagery have been discussed, such as cloud cover and co-registration, although the effects of those variables on model performance has not been tested. The results of this project are limited by not having a comparison to high resolution imagery, and not comparing to state of the art multi-temporal change detection models which may be implicitly preserving spatial details across time. Investigating both of these would give a much better indication of the viability of this method. In terms of time complexity, this method incurs a linear cost in the number of pixels processed compared to using pairs of high resolution images for change detection. The time of training and inference time of the change detection model is the same as using imagery that is natively a 3.3 m resolution instead of super-resolved to 3.3 m. The increased compute time comes from DeepSent inference times. In practical situation, it may also be desirable to re-train DeepSent on imagery within the spatial domain of the change detection dataset used, as out of spatial domain generalisation is still a challenge for change detection models (28). Additional change detection architectures have been not compared, however in theory any bi-temporal change detection model can be put in place of FresUnet with no changes to the pre-processing pipeline. Another question not investigated is the importance of the temporal relation of the leading image to the rest of the images in the time-series stack. In the experiments, image stacks were formed with the first image chronologically being the one being super-resolved. This was done due to limitations in the dataset, however if the last image chronologically was being super-resolved, much more timely change detection could be done, with the only limiting factor being the requirement of the image to super-resolve being cloud free. Additionally, the size and noise limitations of the OSCD dataset could likely be overcome with the use of the SpaceNet 7 dataset, because of its high resolution labels. Notably, SpaceNet 7 labels resemble OpenStreetMap building labels, as both are building polygons made from high resolution imagery. The issue with using OpenStreetMap labels is they are crowdsourced and so will contain noise in the form of incorrect labelling. Training weakly supervised change detection methods is one solution to this (5). If OpenStreetMap data could be used to train change detection models, much larger models could be trained, as both OpenStreetMap, and Sentinel-2 are openly available.

With all of these things together, and supposing good performances compared to high resolution

imagery and multi-temporal change detection methods, this project could be used as an openly available tool for getting estimates of changed buildings anywhere in the world. The method presented could be useful for humanitarian organisations or local governments in deciding whether an area needs further mapping or getting rough urban growth metrics in areas where they are not readily available.

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